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StrataFlux: Memory-Hierarchy-Native Low-Bit LLM Inference

Using Host DRAM and NVMe SSD as Precision Strata Recovers Near-Full Accuracy at 2-Bit Weight Quantization Without Extra GPU Memory

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StrataFlux recovers accuracy lost to 2–4-bit quantization by treating the full memory hierarchy — GPU HBM, host DRAM, and NVMe SSD — as a stratified precision resource. On 2-bit Llama-3.1-8B-Instruct, WikiText-2 perplexity drops from 9.46 to 8.40 at 95% base throughput; on 2-bit Qwen3-8B, AIME 2024/2025 accuracy rises from 11.0% to 47.1% with no additional GPU memory.

AT A GLANCE

Problem: 2-bit quantization drastically reduces GPU memory but collapses accuracy; existing mixed-precision fixes keep all higher-precision weights on-GPU, negating the memory benefit.

Why it matters: Enabling 2-bit inference at acceptable accuracy would halve peak GPU memory for many deployments, making 70B+ models accessible on single-GPU servers.

Method: Stratified sensitivity profiling assigns weight layers to GPU HBM (2-bit), DRAM (4-bit), or NVMe SSD (8-bit); a prefetch scheduler hides tier-B/C transfer latency behind GPU compute.

Result: 2-bit Llama-3.1-8B perplexity 8.40 (vs. 9.46 baseline); 2-bit Qwen3-8B AIME accuracy 47.1% (vs. 11.0%).

Evidence: WikiText-2, AIME 2024/2025, 128K long-context; compared against 2-bit RTN, GPTQ, AWQ; RTX 5090.

The Big Picture

Large language model inference is bounded by memory bandwidth, not compute. Weights must be loaded from GPU memory on every forward pass, and peak GPU memory sets the maximum model size that fits on a single device. Quantization reduces both: 2-bit weights need 16× less memory than FP32 and consume less bandwidth per layer.

The problem is accuracy. At 2 bits, the quantization error per weight is large enough to damage the model's reasoning ability, particularly on tasks that require precise recall of numerical facts or long multi-step chains. Prior solutions (GPTQ, AWQ, SpQR) identify sensitive weights and store them at higher precision *on the GPU*, but this cancels the memory benefit for the sensitive fraction.

StrataFlux reframes the question: the GPU is not the only memory in a modern server. Host DRAM is 8–32× larger and 5–10× slower; NVMe SSD is 10–100× larger still. The sensitive weights that need 4- or 8-bit precision can live in DRAM or SSD and be streamed to GPU only when needed for a specific layer, overlapped with the GPU computation on the previous layer.

Why Existing Approaches Fall Short

Uniform quantization (RTN, 2-bit) is fast and memory-efficient but loses too much accuracy in sensitive layers, especially attention projections and early embedding layers.

Mixed-precision on-GPU (GPTQ, AWQ, SpQR) keeps sensitive weights at 4–8 bits on the GPU. This recovers accuracy but requires extra GPU memory proportional to the number of sensitive parameters, partially defeating the purpose of quantization.

CPU offloading (FlexGen, DejaVu) moves entire layers to CPU for latency-insensitive serving but introduces large latency spikes when layers are paged back in, making throughput unpredictable.

StrataFlux differs from CPU offloading in two ways: (1) it offloads only the *higher-precision copies* of sensitive layers, not the full layer; the base 2-bit copy always stays on GPU for fast path; and (2) a prefetch scheduler computes the access pattern ahead of time and hides DMA transfers behind GPU computation.

Core Mechanism: Three-Tier Precision

Stratum A (GPU HBM): The base quantized weights of all layers at the target bit-width (2–4 bits). Always resident; used for fast-path inference.

Stratum B (host DRAM): 4-bit copies of the top-k% most sensitive layers, identified by calibration. Paged in one block ahead of the layer it corrects.

Stratum C (NVMe SSD): 8-bit copies of an extremely sensitive subset (typically the first and last 2 transformer blocks, plus any layer where calibration error exceeds a threshold). Paged in asynchronously via direct DMA.

For a layer assigned to stratum B or C, the forward pass combines:

$$\mathbf{y} = W_{\text{base}}\mathbf{x} + \Delta W_{\text{tier}}\mathbf{x},$$

where W_{base} is the 2-bit on-GPU weight and $\Delta W_{\text{tier}} =$

$W_{\text{tier}} - W_{\text{base}}$ is the correction loaded from DRAM or SSD. This residual form ensures that if the DMA transfer has not completed (e.g., SSD latency spike), the base path produces an approximate result instead of blocking.

Technical Formulation

Sensitivity of layer l is estimated by the Fisher trace:

$$S_l = \operatorname{tr}(\mathbf{H}_l) \approx \frac{1}{|C|} \sum_{x \in C} \left\| W_l x - \hat{W}_l x \right\|_2^2,$$

where C is a calibration corpus, W_l is the full-precision weight, and $\hat{W}_l$ is its 2-bit quantized version. Layers are ranked by S_l and assigned to strata by two thresholds θ_B and θ_C:

$$\begin{aligned} l &\in \text{Stratum A if } S_l \le \theta_B, \\ l &\in \text{Stratum B if } \theta_B < S_l \le \theta_C, \\ l &\in \text{Stratum C if } S_l > \theta_C. \end{aligned}$$

Thresholds θ_B and θ_C are set by a memory budget constraint: the DRAM and SSD footprints are fixed by the available hardware, and layers are sorted by S_l until the budget is consumed.

Learning or Inference Procedure

StrataFlux requires no fine-tuning; it modifies only the inference memory layout and access pattern. The workflow is:

1. Run calibration pass on 128 calibration sequences to compute S_l for all layers.
2. Assign layers to strata based on thresholds.
3. Quantize and place stratum-B copies in DRAM, stratum-C in SSD.
4. At inference time, the prefetch scheduler issues DMA requests for the next 3 stratum-B layers and 1 stratum-C layer ahead of execution, overlapping transfers with GPU computation.

What the Guarantee Says

No theoretical accuracy bound is provided. The empirical claim is Pareto-dominance over existing mixed-precision methods on the two primary benchmarks: StrataFlux achieves lower perplexity at lower GPU memory than any configuration of GPTQ or AWQ tested. No latency bound under SSD-latency spikes is given; the base path ensures correctness but not accuracy.

Experimental Findings

- **Perplexity:** 2-bit Llama-3.1-8B-Instruct WikiText-2 PPL 8.40 (StrataFlux) vs. 9.46 (2-bit RTN base); 95% base throughput on RTX 5090.
- **Reasoning:** 2-bit Qwen3-8B AIME 2024/2025 accuracy 47.1% (StrataFlux) vs. 11.0% (2-bit RTN); GPU memory unchanged.
- **Long-context:** 3-bit weights + 2-bit KV cache at 128K context matches vLLM 4-bit AWQ + 8-bit KV perplexity and speed at half the GPU memory.
- **Throughput:** Prefetch scheduler achieves 95% of base 2-bit throughput by hiding DRAM latency; SSD accesses visible only when DRAM budget is exhausted.

Ablations and Interpretation

Stratum-C necessity: Removing stratum C (SSD tier) and assigning all stratum-C layers to stratum B (DRAM at 4-bit) increases DRAM footprint by 18% at no accuracy change, confirming that the SSD tier covers only a small but critical fraction of sensitive layers.

Residual form: Replacing the residual correction ΔW_{tier} with a full replacement (always load W_{tier} on this path) reduces throughput by 12% because the base 2-bit GPU path is always bypassed, even when DMA is on time.

Calibration size: 32 calibration sequences achieve 94% of the sensitivity ordering quality of 512 sequences; 128 sequences reach 99%.

Model size: Relative perplexity improvement is larger for 8B-class models (1.1 PPL points) than for 70B-class models (0.4 PPL points), reflecting that smaller models are more sensitive to 2-bit quantization and benefit more from stratum correction.

Reference

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